

Family Name/
Surname: _____

First Name/
Given Name: _____

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Venue: _____

Seat Number: _____

Mathematics and Statistics**TEST**

Mid-semester Test, 2025

EMTH210-25S1 (C) Engineering Mathematics 2Time allowed: **1.5 HOURS**Number of questions: **4**Number of pages: **20****Instructions to Students:**

- Answer all 4 questions.
- There is a total of 80 marks.
- Show all working.
- This test is restricted book. The formula sheet will be supplied separately.
- Communication with others by any means is not permitted.
- No calculators are permitted.

**For Examiner Use
Only**

Question	Mark
Q1 /28	
Q2 /22	
Q3 /15	
Q4 /15	
Total /80	

Questions Start on Page 3

1. (28 marks)

(a) Consider the derivative $\left(\frac{\partial G}{\partial F}\right)_x$ in the context of the equations

$$F = x^2 e^{3xy} \quad \text{and} \quad G = \sin(y) \cos(2x).$$

List the dependent and independent variables and use implicit differentiation to find

$$\left(\frac{\partial G}{\partial F}\right)_x.$$

(5 marks)

(b) Evaluate the following line integral, I , in the (x, y) -plane:

$$I = \int_C 3 \sin(x) dx - y dy$$

Here, the curve $\mathcal{C}$ is the curve $y = x^2$ from $(x, y) = (0, 0)$ to $(x, y) = (\pi, \pi^2)$.

You must parameterise in terms of t .

(6 marks)

(c) i. State the test to determine whether $P dx + Q dy$ is exact. (1 mark)

ii. Use the test to determine whether $(x + y^2) dx + (x^2 + y) dy$ is exact. (2 marks)

(d) i. Find a function f such that its total differential is the exact differential

$$df = ye^{xy} dx + xe^{xy} dy.$$

(4 marks)

ii. Use your function from part i to evaluate the integral

$$\int_{(1,0)}^{(2,2)} ye^{xy} dx + xe^{xy} dy.$$

(2 marks)

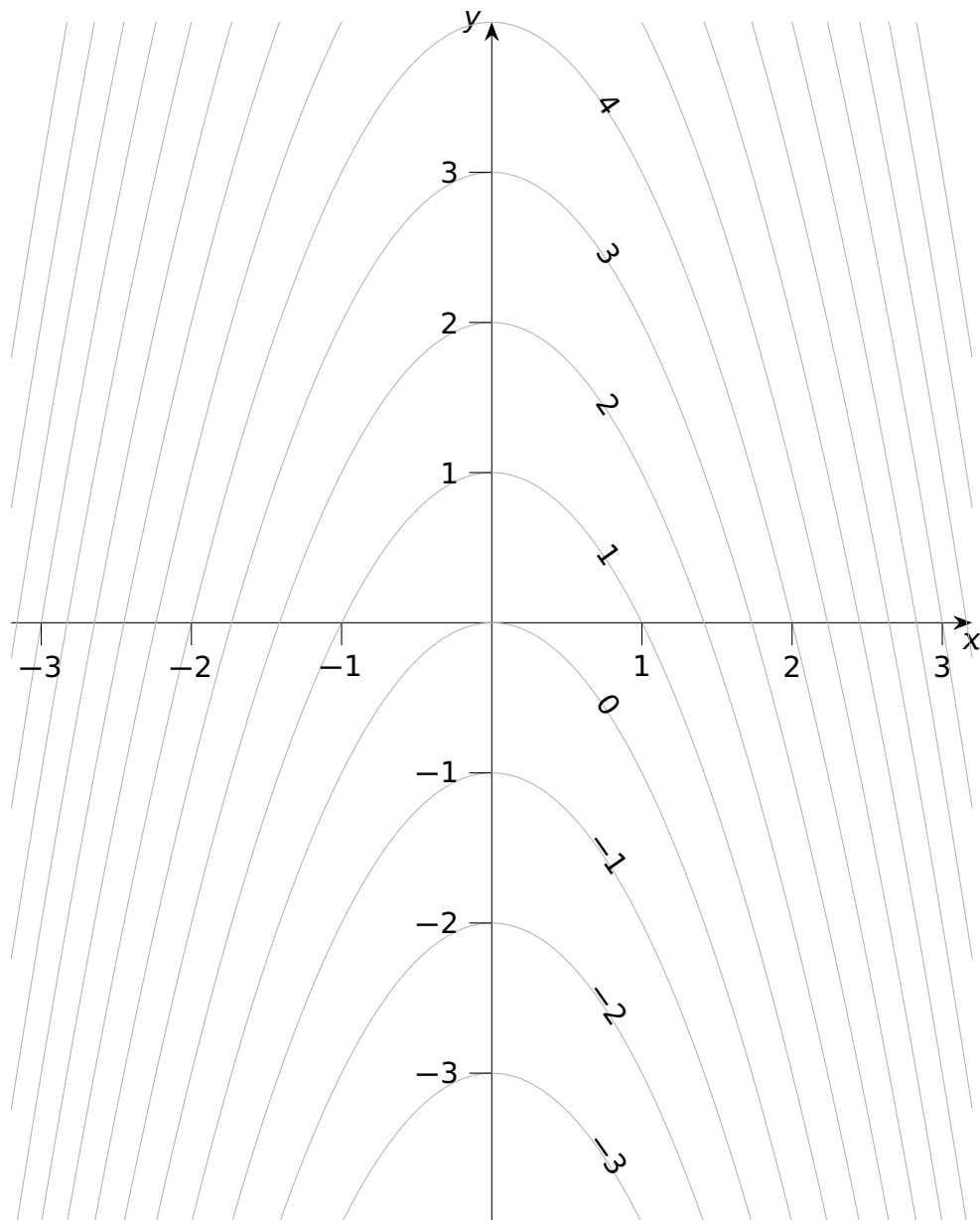
- (e) i. Calculate the directional derivative of $f(x, y) = x^2y^2$ in the direction of the unit vector $\underline{\hat{u}} = \begin{bmatrix} u \\ v \end{bmatrix}$ at the point $(1, 2)$. (4 marks)

- ii. Find a unit vector $\underline{\hat{u}}$ that describes a direction in which f is stationary at the point $(1, 2)$. (4 marks)

Spare page for extra working. Please write the question number(s).

2. (22 marks)

- (a) Consider the function $f(x, y) = x^2 + y$ on the constraint curve $c(x, y) = 36 - 9x^2 - 4y^2 = 0$. The plot below shows level curves of f . On the same plot, do the following.
- Sketch the constraint curve. (2 marks)
 - Mark the approximate locations of the constrained critical points. (2 marks)
 - Sketch (and label) the directions of ∇f and ∇c at the constrained critical points. (2 marks)
 - Identify the critical points as constrained relative maximums or minimums. (2 marks)



(There is a spare plot on page 11.)

- v. Use Lagrange multipliers to find the critical points of $f(x, y) = x^2 + y$ subject to the constraint $c(x, y) = 36 - 9x^2 - 4y^2 = 0$. (8 marks)

(b) Consider the vector field $\underline{\mathbf{f}} = \begin{bmatrix} x^2yz \\ xz \\ y \end{bmatrix}$.

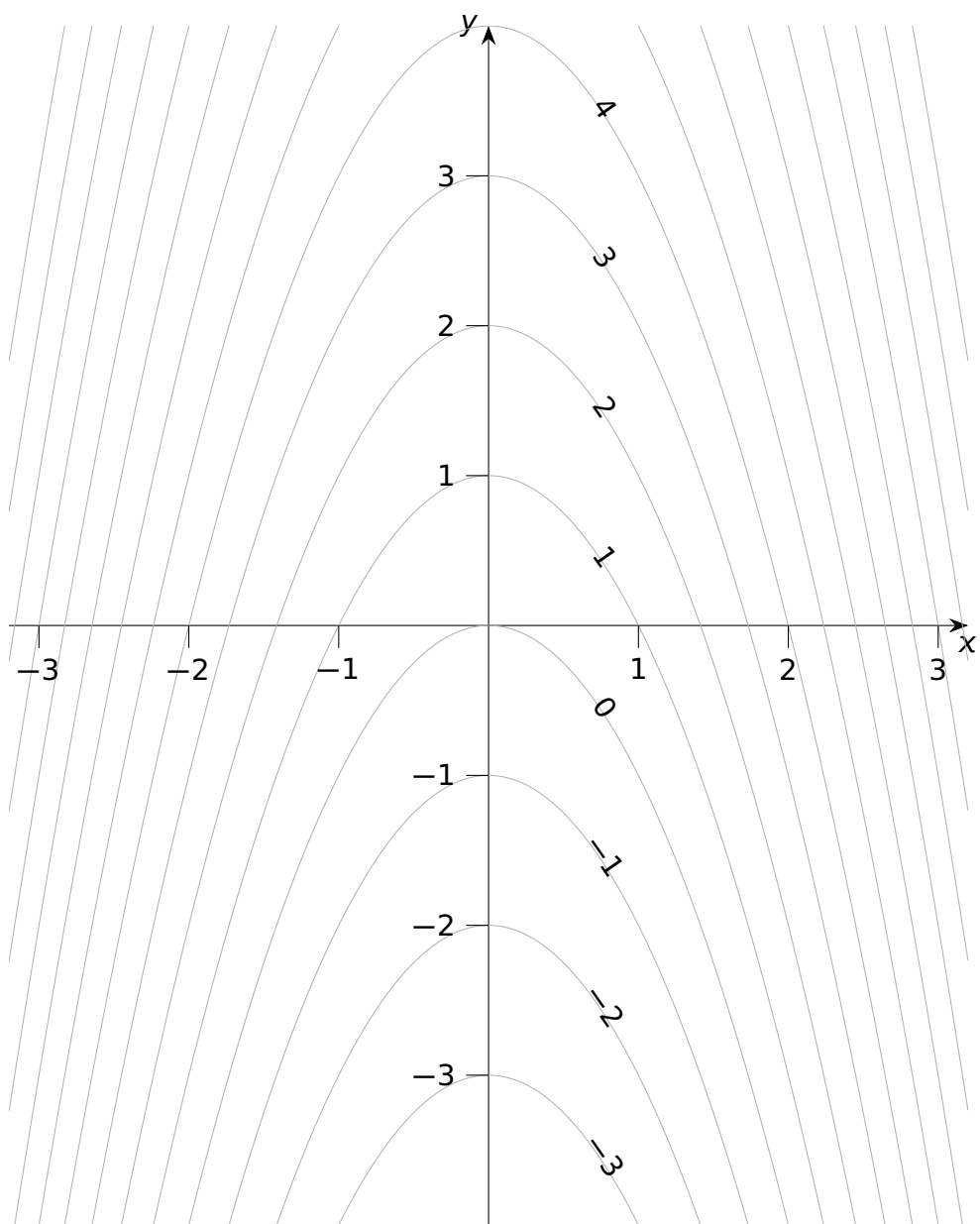
i. Calculate the curl and divergence of $\underline{\mathbf{f}}$.

(4 marks)

ii. Find a point at which $\underline{\mathbf{f}}$ is irrotational.

(2 marks)

(Spare plot for 2a.)



3. (15 marks)

(a) Find the solution to the differential equation

$$\frac{d^2y}{dt^2} + 2\frac{dy}{dt} + y = 0$$

which satisfies the boundary conditions $y(0) = 0$ and $y(4) = 4$.

(5 marks)

- (b) Using the method of undetermined coefficients, find the general solution to the differential equation

$$\frac{d^2y}{dt^2} + 4\frac{dy}{dt} + 5y = \sin(t) + 5t.$$

No marks will be awarded for solutions by any other means.

(8 marks)

(3b continued...)

- (c) Consider the differential equation $y'' + 4y = 3\cos(2t)$, which has complementary function $y_c = C_1 \cos(2t) + C_2 \sin(2t)$. Without doing any calculations, what effect do you expect to see in the general solution's behaviour, and why? (2 marks)

Spare page for extra working. Please write the question number(s).

4. (15 marks) A 2π periodic function $f(x)$ is defined by

$$f(x) = \begin{cases} -\pi - x & \text{if } -\pi < x < 0 \\ \pi - x & \text{if } 0 < x < \pi, \end{cases}$$

and $f(x) = f(x + 2\pi)$ elsewhere.

(a) Sketch two periods of $f(x)$.

(2 marks)

(b) Is this function odd, even, or neither? Justify your answer.

(2 marks)

(c) Find the Fourier series for $f(x)$.

(7 marks)

(part (c) continued...)

- (d) Using your answer to part (c) of this question, find the Fourier series of the function $g(t)$ defined by

$$g(t) = \begin{cases} -\pi - \pi t & \text{if } -1 < t < 0 \\ \pi - \pi t & \text{if } 0 < t < 1, \end{cases}$$

and $g(t) = g(t + 2)$ elsewhere. No further integration is required. (2 marks)

- (e) Find the fundamental frequency of the function

$$h(x) = 4 + \sin(3\pi x) + \cos(4\pi x).$$

(2 marks)

End of Test

Spare page for extra working. Please write the question number(s).